

■# PAPER<i>354 — D</i>Universe 5th Factor: Spatial Curvature Completion of the 4-Factor Chain (PAPER_296)

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■# PAPER354 — DUniverse 5th Factor: Spatial Curvature Completion of the 4-Factor Chain (PAPER_296)

Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF 5th factor spatial curvature term for *Duniverse*; *completes PAPER296 chain* **Author:** Daniel T. Murphy

<!-- UQFF constants: $\kappa = 5.0\text{e-4 day}^{-1}$, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e1 TeV}$ -->

Abstract

PAPER296 established a 4-factor chain for the UQFF Universe expansion parameter *Duniverse*. This paper adds the mandatory 5th factor: a spatial curvature correction $(1 + k \cdot r_c^2)$, where k is the curvature constant and r_c is the Friedmann comoving curvature radius. The complete 5-factor *Duniverse* is now: $D_{\text{universe}} = [4 \text{ prior factors}] \times (1 + k \cdot r_c^2)$. For a flat universe ($k = 0$), the 5th factor = 1 and PAPER296 is recovered. For non-flat models, this term accounts for the deviation of cosmic spatial geometry from the Minkowski approximation used in earlier UQFF distance calculations.

2. Core Physics

2.1 Complete 5-Factor *D_universe*

$$D_{\text{universe}} = D_1 \cdot D_2 \cdot D_3 \cdot D_4 \cdot (1 + k_{\text{curv}} \cdot r_c^2)$$

where factors *D1 through D4* were established in PAPER_296 (Session 91) and the new 5th factor is:

$$D_5 = 1 + k_{\text{curv}} \cdot r_c^2$$

2.2 Curvature Parameter k

The Friedmann equation curvature:

$$k_{\text{curv}} = \frac{(H_0^2 / c^2)(\Omega_{\text{total}} - 1)}{1}$$

For the Planck 2018 constraint $\Omega_{\text{total}} = 1.0007 \pm 0.0019$:

$$k_{\text{curv}} \approx 0.0007 \cdot \frac{H_0^2}{c^2} \approx 5.3 \times 10^{-54} \text{ m}^{-2}$$

2.3 Curvature Correction at Cosmological Scale

At $r_c = \text{Hubble radius}$ ($R_H = c/H_0 \approx 1.37 \times 10^{26} \text{ m}$):

$$D_5 = 1 + 5.3 \times 10^{-54} \cdot (1.37 \times 10^{26})^2 = 1 + 5.3 \times 10^{-54} \cdot 1.88 \times 10^{52}$$

$$D_5 = 1 + 0.001 = 1.001$$

A 0.1% correction — detectable by next-generation CMB experiments (e.g., CMB-S4, LiteBIRD).

2.4 Near-Flat Expansion Series

For small curvature ($k_{\text{curv}} \cdot r_c^2 \ll 1$):

$$D_5 \approx 1 + k_{\text{curv}} r_c^2 - \frac{(k_{\text{curv}} r_c^2)^2}{2} + \dots$$

The leading correction is linear in both k and r_c^2 .

3. Key Values

Quantity	Formula	Value
k_{curv}	Planck 2018 constraint	$\sim 5.3 \times 10^{-7} \text{ m}^{-2}$
r_c (Hubble radius)	c/H_0	$1.37 \times 10^{26} \text{ m}$
D_5 (Hubble scale)	$1 + k \cdot r_c^2$	1.001
D_5 (flat limit)	$k = 0$	1.000
PAPER_296 factors	$D1 \times D2 \times D3 \times D4$	Previously computed

4. Physical Significance

The 5th factor completes the UQFF *Duniverse chain*, which now accounts for: (1) vacuum buoyancy scale factor, (2) string rotation expansion term, (3) Hubble flow scale, (4) charge-reactivity expansion coupling, and (5) spatial curvature geometry. The chain $D_{\text{universe}} = D1 \times D2 \times D3 \times D4 \times D5$ represents the most complete UQFF treatment of cosmic expansion parameters. The 0.1% curvature correction at Hubble scale sets the signal size for CMB observational tests: future CMB-S4 measurements of the spatial curvature power spectrum should detect $D5$ deviations from unity at the $\sim 0.05\%$ level.

5. Deduplication Note

****vs. PAPER_296:**** PAPER_296 derived the 4-factor chain; PAPER_354 adds the mandatory spatial curvature 5th factor.
Unique: The $(1 + k \cdot r_c^2)$ form is new — no earlier UQFF paper included spatial curvature directly in *Duniverse*.

6. Classification

Physics Territory: FIRST UQFF *Duniverse spatial curvature 5th factor*; completes PAPER296 four-factor chain **Scale:** Cosmological (Hubble radius; universal) **CP Implementation:** `DUniverseSpatialCurvatureFifthFactorCalculator` (CondensedPhysics3.py, Session 96)